

LESSON 2

NOT, AND, OR

Name: _____ Date: _____

Compound statements

Simple statements can be combined with **logical connectives**. Think of them as glue between ideas. We use symbols so long sentences stay readable.

NOT · $\neg p$

"It is not the case that p ."

Flips $T \leftrightarrow F$. Also written " $\sim p$ " in some books.

AND · $p \wedge q$

" p and q ."

True only when **both** parts are true.

OR · $p \vee q$

" p or q " (inclusive)

True when **at least one** part is true (including both).

NOT in detail

If p = "The door is locked" is **true**, then $\neg p$ = "The door is not locked" is **false**.

DOUBLE NEGATION

Negating twice returns to the original truth: $\neg(\neg p)$ has the same truth value as p .

"It is not true that I did not finish" $\rightarrow$ means I *did* finish (clumsy English, clear logic).

AND in detail

AND is strict. One false part sinks the whole compound.

- "I have a ticket **and** I have ID" $\rightarrow$ need both to enter.
- $\text{True} \wedge \text{False} \rightarrow \text{False}$
- $\text{False} \wedge \text{False} \rightarrow \text{False}$

OR in detail — inclusive vs everyday "pick one"

In logic, OR is **inclusive**: both parts can be true and OR is still true.

INCLUSIVE OR

"You may have soup or salad" at a buffet often means you *can* take both. Logical OR matches that generous reading.

Exclusive OR ("exactly one, not both") is a different connective — we write XOR or " $p \oplus q$ " in advanced courses. Unless a problem says "exactly one," assume inclusive OR.

Parentheses matter

Like arithmetic, grouping matters: $(p \wedge q) \vee r$ is not the same as $p \wedge (q \vee r)$. Always write parentheses when mixing connectives.

WORKED EXAMPLE

Let p = "I finished homework." (T) · q = "I studied for the quiz." (F) · r = "I cleaned my room." (T)

- $\neg q \rightarrow$ True
- $p \wedge q \rightarrow$ False
- $p \vee q \rightarrow$ True
- $(p \wedge q) \vee r \rightarrow (F) \vee T \rightarrow$ True
- $p \wedge (q \vee r) \rightarrow T \wedge (T) \rightarrow$ True

MEMORY TRICK

AND is picky (all T). OR is generous (one T enough). NOT is a light switch.

Practice — Lesson 2

2.1 WARM-UP

Let p be **true** and q be **false**. Find the truth value:

1. $\neg p$

2. $\neg q$

3. $p \wedge q$

4. $p \vee q$

T OR F FOR EACH

2.2 WARM-UP

Translate into symbols. Define p and q first.

1. The store is open and I have money.

2. It is not raining.

3. I will take the bus or I will walk.

DEFINITIONS + SYMBOLIC FORM

2.3 STANDARD

$p = \mathbf{T}$, $q = \mathbf{T}$, $r = \mathbf{F}$. Evaluate:

1. $\neg(p \vee r)$
2. $(p \wedge r) \vee q$
3. $\neg p \vee (q \wedge r)$

SHOW INTERMEDIATE STEPS

2.4 STANDARD

Write each English sentence in logic symbols (define letters).

1. I am not tired and I am not hungry.
2. Either the red team or the blue team wins (or both tie — use inclusive OR).
3. It is false that the library is closed and the café is closed.

SYMBOL VERSIONS

2.5 STANDARD

For each pair, explain in one sentence whether they mean the same thing:

1. “You need a password and a username.” vs “You need a username and a password.”
2. “Pick coffee or tea (choose only one).” vs logical OR without the “only one” rule.

SAME OR DIFFERENT? WHY?

2.6 CHALLENGE

p, q, r are all unknown. Find **one assignment** of T/F to p, q, r that makes the whole compound **true**, and one that makes it **false**:

$$(p \vee \neg q) \wedge (q \vee r) \wedge \neg(p \wedge r)$$

TRUE ASSIGNMENT / FALSE ASSIGNMENT + BRIEF CHECK

2.7 CHALLENGE

Write a real-life rule (school, sport, or game) using at least two connectives. Give definitions for your letters and explain when the rule passes vs fails.

RULE + ANALYSIS

